Seminar: Statistics for Data Science: Group **C**

Academic year 2025/2026

1 Data about students

Names of three students
Date of presentation : TBA
Data or sources : grades in mathematics and language in a sample of high-school students

2 Content - Student performance

Educational datasets provide detailed information on studentsâ€™ demographic characteristics, family background, study habits, and academic outcomes. Student performance is influenced by multiple factors, and its evolution over time can offer additional insights into learning processes.

The aim of this seminar is to investigate the association between studentsâ€™ final grade (variable `G3`) and the available explanatory variables, taking into account both the temporal structure of grades and the presence of multiple subjects.

The datasets that you will use contain information on secondary school students in Portugal, for two subjects: Mathematics and Portuguese language. For each student, grades are recorded at three time points (`G1`, `G2`, `G3`), together with a wide range of demographic, social, and academic variables.

Most of the data refer to different students, but a subset of students is present in both datasets. These students can be identified by matching a set of common attributes.

Data or sources

Download the dataset from the official UCI repository:

`https://archive.ics.uci.edu/static/public/320/student+performance.zip`

After downloading:

- extract the ZIP file
- locate the files `student-mat.csv` and `student-por.csv`
- use one or both files for the analysis

Import the data into R. For example:

```
setwd("C:/your/path/to/student+performance")

d1 <- read.table("student-mat.csv", sep=";", header=TRUE)
d2 <- read.table("student-por.csv", sep=";", header=TRUE)

head(d1)
str(d1)

head(d2)
str(d2)
```

Data content and data management

Read carefully the definition of the variables in the `student.txt` file included in the dataset.

Note that some of the variables are provided in transformed form, obtained by scaling them (dividing by their maximum value, hint: focus on the variables related to weather conditions).

Transform these variables back to their original scale before proceeding with the analysis, and clearly describe how the transformation was performed.

Use the transformed variables in the analysis and clearly describe the transformations.

Some variables are categorical but coded as numerical values (for example `season`, `weathersit`, `weekday`, `mnth`).

Recode these variables by assigning meaningful labels (e.g. factor variables in R), so that the results of the analysis are clearly interpretable.

Data content and data management

Read carefully the definition of the variables in the `student.txt` file included in the dataset.

The datasets include a mix of numerical, categorical, and ordinal variables.

Some variables are categorical but coded as strings (for example `school`, `sex`, `Mjob`, `Fjob`, `reason`, `guardian`).

Recode these variables by assigning meaningful labels (e.g. factor variables in R), so that the results of the analysis are clearly interpretable.

Some variables are ordinal (for example `Medu`, `Fedu`, `studytime`, `health`, `goout`). Consider how these variables should be treated in the analysis and clearly justify your choices.

If both datasets are used, students present in both datasets can be identified by merging the datasets using common attributes. For example:

```
d1 <- read.table("student-mat.csv", sep=";", header=TRUE)
d2 <- read.table("student-por.csv", sep=";", header=TRUE)

d3 <- merge(d1, d2,
  by=c("school","sex","age","address","famsize","Pstatus",
       "Medu","Fedu","Mjob","Fjob","reason","nursery","internet"))

nrow(d3)  # number of matched students
```

Clearly describe and justify any data preprocessing steps, including recoding of variables and, if applicable, the merging procedure.

Definition of the research question and of the model for the primary analysis

The aim of the primary analysis is to investigate the association between the studentsâ€™ final grade (`G3`) and the available explanatory variables.

Define clearly your research question in terms of the outcome and the explanatory variables of interest.

The analysis can be performed on one of the two datasets (Mathematics or Portuguese), or on both datasets separately.

Address the research question using a linear regression model, with `G3` as the outcome variable.

Discuss whether early grades (`G1`, `G2`) should be included as explanatory variables, and justify your choice.

Perform an Initial Data Analysis (IDA), guided by your research question, and briefly present your findings.

Use the checklist for regression models proposed in Heinze et al. ("Regression without regrets – initial data analysis is a prerequisite for multivariable regression", BMC Medical Research Methodology, 2024), available at:

`https://doi.org/10.1186/s12874-024-02294-3`

Carefully consider missing data, and aspects related to the univariate and multivariate distributions of the variables.

Explicitly state if the IDA findings determined changes in the analysis strategy (and describe the changes, if any).

Address the research question fitting a linear regression model.

Summarize the main results and provide a clear interpretation of the estimated associations in terms of the original variables. Clearly report the final model specification.

Focus on the practical meaning of the results (e.g. how changes in explanatory variables such as study time, absences, or family background are associated with changes in the final grade `G3`), not only on statistical significance.

If both datasets are analyzed, compare the results obtained for Mathematics and Portuguese, highlighting similarities and differences in the estimated associations.

Evaluate the appropriateness of the model using model diagnostics.

Assess whether the assumptions of the linear regression model are reasonable. In particular, consider aspects such as:

- the distribution of residuals
- possible patterns in residuals (e.g. non-linearity, heteroskedasticity)
- the presence of influential observations

Present the findings with tables and graphical displays.

Investigate possible non-linear relationships between the explanatory variables and the outcome. - IF NOT DONE IN THE PREVIOUS PARTS OF THE SEMINAR

In particular, focus on the relationship between selected numerical or ordinal variables (for example `studytime`, `absences`, `age`, or variables related to lifestyle) and the final grade `G3`.

Explore whether a linear effect is appropriate, and consider alternative specifications (for example, including a quadratic term, grouping categories, or using other transformations).

Present the results graphically and comment on how the relationship differs from a simple linear effect.

Discuss how accounting for non-linearity affects the interpretation of the results and the overall fit of the model.

Provide a brief critical discussion of your findings.

Comment on the limitations of your analysis and possible improvements.

Discuss also the use of the fitted model for prediction. In particular, consider the case where a model is fitted using one dataset (e.g. Mathematics) and then used to predict the final grade in the other dataset (e.g. Portuguese). Discuss the potential limitations and issues of this approach, and what factors might affect the validity of the predictions.

In your discussion, consider in particular:

- the role of early grades (`G1`, `G2`) and whether they should be treated as explanatory variables or as part of the same outcome process;
- the temporal structure of the data (repeated measurements of grades for the same student) and its implications for the analysis;
- differences between subjects and whether a single model can adequately describe both;
- the subset of students present in both datasets and possible selection effects arising from the merging procedure.

Length of written report (to be turned in two days before the presentation): at most 5 pages of text and 5 pages with figures/text.

Length of presentation (no need to submit the file): 20 minutes + discussion.

Good luck!